

# BOM requirements

What the schematic has to buy, and the simulation that says so.

Every row here exists because a block in `sim/blocks/` failed without it. The rule this file follows is the same one the netlists follow: **a figure nobody has read in a datasheet does not become a design number**. So these are requirements stated as specifiable, purchasable properties — tempco bands, ESR bands, on-resistance ceilings — not manufacturer part numbers. Naming a part is a separate step, and it needs a retrieved datasheet.

Where a part number *does* appear below, it is because it was checked.

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## Input protection

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### TVS — transient suppressor, battery input

SMCJ43CA. Bidirectional, 43 V standoff, 1500 W package.

Both halves of that were forced by simulation, not chosen:

- **43 V standoff** is the smallest standard bin above the 40 V clamped load dump. Below it the part conducts during a *normal* ISO 7637-2 pulse 5b event rather than a fault — which is what `load_dump.cir` was failing on, not "the TVS is too small."
- **1500 W (SMC) package, not 600 W (SMB)**. At the SMB size, pulse 2a clamps at 88.6 V against the buck's 100 V absolute maximum — 11% margin. The SMC part has roughly half the dynamic resistance at the same standoff and clamps at 73.3 V, 1.37×.

Checked by `load_dump.cir` (0.000 A, 0.000 J through the TVS) and `transient_clamp.cir`.

### NEG-CLAMP — negative active clamp, buck VIN

N-channel MOSFET,  $V_{ds} \geq 40 \text{ V}$ ,  $R_{ds(on)} \leq 10 \text{ m}\Omega$  at  $V_{gs} = 4.5 \text{ V}$ , source to ground, drain to VIN, driven by a comparator sensing VIN against a  $-0.1 \text{ V}$  threshold. **In parallel with** the 60 V / 20 A clamp Schottky, which stays as the instant passive backstop.

The requirement is really an *effective clamp impedance*, and it is worth stating as one: the LM5164's VIN absolute maximum is  $-0.3 \text{ V}$ , pulse 1 delivers 150 V through  $10 \text{ }\Omega$ , so

$$R_{\text{clamp}} \leq 0.3 \text{ V} / 15 \text{ A} = 20 \text{ m}\Omega, \text{ junction drop included}$$

No junction device meets that, and the reason is structural rather than a sizing miss. Decomposing the Schottky's own drop at 15 A:

| term                                     | value   |
|------------------------------------------|---------|
| junction, $N \cdot V_t \cdot \ln(I/I_s)$ | 0.367 V |
| bulk, $I \cdot R_s$                      | 0.180 V |
| total                                    | 0.547 V |

Only the bulk term falls when parts are paralleled; the junction term is logarithmic in current and barely moves. Four Schottkys in parallel still leave 0.375 V. A FET in its ohmic region has no junction offset at all — 15 A through 5 mΩ is 75 mV.

Checked by `negative_pulses.cir`: buck VIN reaches -0.068 V on pulse 1 (4.4× inside the rating) and -0.027 V on pulse 3a (11×), against -0.528 V and -0.393 V with the Schottky alone.

**Open, and bounded:** the comparator's propagation delay is not modelled. VIN sits at the Schottky's -0.53 V for exactly that long before the clamp engages. What changed is the size of the question — it was "is a 2 ms excursion acceptable," and it is now "is a sub-microsecond one," which a comparator datasheet answers.

## Clamp Schottky

60 V / 20 A class. It carries what the TVS used to: **13.58 A peak, 14.5 mJ** on pulse 1, measured at its own ammeter in `negative_pulses.cir` rather than inferred from 150 V / 10 Ω.

## Reverse-battery FET

**100 V class** — its third revision, raised each time the clamp moved. `transient_clamp.cir` now carries an explicit check stating this downstream requirement, so the next move does not depend on someone reading three files and noticing.

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## Power

### PDN-BULK — 3V3\_MCU bulk capacitor

Polymer aluminium electrolytic or polymer tantalum, 150 μF, specified by an **ESR band of 20–50 mΩ across -40...+125 °C** — not by a 25 °C typical figure.

Three things about that are load-bearing:

- **The part class changes, not the value.** A wet aluminium or tantalum electrolyte loses conductivity as it freezes, roughly 4× ESR at -40 °C, which puts the PDN peak at 197 mΩ against a 100 mΩ target. A solid

polymer electrolyte has no such mechanism.

- **The cold number is what gets specified.** A supplier's 25 °C ESR does not answer this block's question, and asking for it is how the original corner got in.
- **A band, not a ceiling.** Bulk ESR is also what damps the regulator-against-bulk resonance. At 20 mΩ the peak climbs back to 92 mΩ, 8 mΩ under target. "ESR ≤ 50 mΩ" would be satisfied by a 5 mΩ part that fails this block.

The 100 mΩ target is derived — 50 mV allowed ripple over a 0.5 A transient step — not a round convention. Checked by `mcu_pdn.cir`.

### SENSOR-PTC — 5 V sensor rail, one per sensor group

Resettable PTC, hold current ≥ 200 mA, R25 = 3.0 Ω nominal.

Sized against the ceiling rather than guessed. The polyfuse has to be what clears a shorted harness, not the 5 V regulator's own current limit, so fault current must stay under 3.0 A:

$$\begin{aligned} I < 3.0 \text{ A} &\Rightarrow R_{\text{src}} + R_{\text{ptc}} > 5/3 = 1.667 \text{ } \Omega \\ \text{corner} &= R_{25} \times 0.8 \text{ (cold)} \times 0.7 \text{ (-30\% spread)} = 0.56 \times R_{25} \\ &\Rightarrow R_{25} > 2.78 \text{ } \Omega \end{aligned}$$

At the same stacked -40 °C + -30% corner the specified part draws 2.88 A, 0.12 A under the ceiling. The larger series resistance was re-checked against the rest of the block rather than assumed harmless: healthy-group voltage holds at 4.69 V against a 4.5 V floor, and nominal fault current stays inside the 0.3–3.0 A window the polyfuse needs to trip on.

Checked by `sensor_rail.cir`.

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## Drivers

### MU-ISENSE — metering unit current sense, ECU pin 88 return

50 mΩ shunt on the low side, into **either** a current-sense amplifier feeding an ADC channel **or** a gate driver with an integrated current-regulation comparator.

The amplifier is not optional: 50 mΩ at the 0.675 A setpoint is 33.75 mV, too small to take straight to the S32K148's ADC at useful resolution.

This one is not a part-selection fix at all — it is an architecture change, and it is the only failing corner in the suite that no component choice could have answered. The coil is copper, copper moves 0.39%/°C, and over -40...+125 °C its resistance goes 7.39 → 13.82 Ω. At a fixed 50% duty the mean current follows it: **0.862 A cold against 0.462 A hot**, +27% and -32% around the setpoint. There is no better resistor to buy.

A hysteretic current regulator holds 0.679 / 0.679 / 0.683 A at -40 / 25 / +125 °C. **Closed loop only works while the regulator has duty left** — at the hot corner the coil can draw at most  $13.5/13.82 = 0.977$  A at 100% duty, so the setpoint sits at 69% of the ceiling. That bound is checked, because a regulator out of headroom degrades silently back to open-loop behaviour rather than reporting anything.

Checked by `metering_unit_pwm.cir`.

### **EGR-DRIVER** — EGR H-bridge

A bridge driver with **internal DIR/PWM decode** — one logic input pair that cannot express both legs of a diagonal conducting at once.

Two logic lines wired straight to two diagonal switch pairs short the bridge: with both inputs asserted it draws 270 A. Under a decoded interface the same input combination draws 4.35 A, the motor's own stall current. The fault state stops existing rather than being avoided by firmware convention.

Current capability: headroom over a 2.6–6.4 A stall.

No part is named. None has been checked against a retrieved datasheet. What the schematic needs from the BOM is the interface type, the current limiting, and that headroom.

Checked by `egr_hbridge.cir`, which keeps the naive wiring as pinned evidence of why the requirement exists.

### **GATE-PULLDOWN** — fail-safe gate pulldowns, six driver gates

470 Ω, gate to source, on every driver gate.

The largest standard value that still holds a gate below  $V_{gs(th)} = 1.0$  V against Miller coupling at  $C_{rss} = 500$  pF. The commonly-reached-for 10 kΩ fails even at the *low* end of the memo's own  $C_{rss}$  envelope — 1.155 V at 50 pF already exceeds the threshold.

Cost, stated because it is real: 21.3 mA per gate held at 10 V, 1.28 W across all six.

Checked by `supervisor.cir`.

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## Signal

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### **CAN-TERM** — CAN split termination, $R_{ah}$ / $R_{al}$

60.0 Ω **thin-film**, initial tolerance  $\pm 0.1\%$  or better, tempco  $\pm 50$  ppm/°C or better.

Ordinary standard-grade thick-film runs 100–200 ppm/°C. At 200 ppm/°C over the full –40/+125 °C span, both resistors shifting together, the differential impedance reaches 123.95 Ω against J1939's own 120 Ω — and 120 Ω here is what the standard specifies, not a round convention, so missing it is a real finding.

The contrast is a part class, with no meaningful cost or availability penalty at this quantity. With both mechanisms stacked the adverse way ( $60 \times 1.00825 \times 1.001 = 60.5555 \Omega$ ) the specified part lands at 121.10 Ω, inside the  $\pm 2 \Omega$  window with 0.90 Ω to spare. Checked at the **worst case**, not the nominal — so passing means the requirement is sufficient, not merely that it helps.

Checked by `can_termination.cir`.

### VR-CLAMP — crank input clamp diodes, four off

Reverse leakage  $\leq 1 \mu\text{A}$  at 125 °C. Forward drop is explicitly *not* a requirement.

This inverts the rule every earlier clamp on this board was chosen by. The negative-clamp Schottky's 0.547 V at 15 A decomposes into a junction term and a bulk term, and that decomposition is what ruled out paralleling junction devices for the power path. These four diodes conduct only on a fault, into a 4.7 kΩ series resistor that is already doing the limiting — what matters is what they do when they are off.

A reverse-biased diode's leakage flows out of the 4.489 kΩ Thévenin at the comparator input and appears there as an offset voltage, against a hysteresis window of  $\pm 198 \text{ mV}$ :

| Class                            | Leakage at 125 °C                                                                   | Offset             | Against the window              |
|----------------------------------|-------------------------------------------------------------------------------------|--------------------|---------------------------------|
| BAT54S-class Schottky            | $\sim 200 \mu\text{A}$ ( $\approx 2 \mu\text{A}$ at 25 °C, $\sim 100\times$ hotter) | 0.90 V             | 4.5× the whole window — latches |
| BAV199-class low-leakage silicon | $< 1 \mu\text{A}$ (5 nA max at 25 °C)                                               | $< 4.5 \text{ mV}$ | 2% of the window                |

The two diodes on a leg are reverse-biased symmetrically by `VR_BIAS`, and the legs are matched, so leakages partly cancel and what survives is partly common-mode. That is not a thing to defend a speed input on, so the requirement is stated instead.

`vr_conditioner.cir` does **not** model this — its comparator is an ideal switch with no input network, so no clamp, no leakage and no offset exist in it. Found while drawing `hw/speed_inputs.kicad_sch`.

The same tag covers the **differential TVS across the pair**: bidirectional, standoff  $\geq 48 \text{ V}$ . The standoff has to clear the *signal*, not just the rail — VR amplitude is proportional to speed, about 20 V peak at 1500 rpm, so a 24

V part would clip a runaway engine's own crank signal at the one moment the ECU most needs to keep counting teeth. 48 V clears 3600 rpm.

**VR-COMP — crank zero-cross comparator**

Single 3.3 V supply, **push-pull** rail-to-rail output, propagation delay  $\leq 1\ \mu\text{s}$ . Input common-mode range: **mid-rail only**.

Drawn as `COMPARATOR_GENERIC` — placeholder pins, no footprint — because no part is chosen. Three of those four are unusually easy, deliberately:

- The output swing *is* the hysteresis window ( $3.3\text{ V} \times 4489 / (4489 + 33000) = 0.395\text{ V}$ ), so an open-drain part would make the window depend on its pull-up value and its  $V_{OL}$ .
- Biasing the floating VR coil to `VR_BIAS` means the inputs never leave mid-rail, so **no common-mode range including ground is needed** — the usual hard constraint on a zero-cross detector, avoided by biasing the sensor rather than level-shifting it.
- $1\ \mu\text{s}$  is 0.009 crank degrees at 1500 rpm on a 60-tooth wheel.
- Input offset voltage and its tempco are non-issues: 10 mV is 5% of the window.

Checked by `vr_conditioner.cir`, which establishes the topology (a fixed 5 V threshold produces edges at 1500 rpm and **none** at cranking speed) and the  $\pm 0.2\text{ V}$  window the E24 network above lands on.

**BOOST-BLEED — injector boost reservoir bleed**

22 k $\Omega$ , 1 W, across the 47  $\mu\text{F}$  / 100 V reservoir.

The charger regulates from **one side only**: `boost_converter.cir`'s `Bchg` adds 50 mA while the rail is below setpoint and adds nothing above it. Nothing else in the circuit can remove charge, so any excursion above 100 V is permanent. `injector_turnoff.cir` measured the mechanism that produces one — a hold-current cutoff, whose energy came from the **battery** through the high side's diode-OR and so has no boost-side draw to net against — and its RESULT NOTE asked for this resistor by name.

Sizing is not a duty-cycle argument; it is "one event is gone before the next cylinder fires":

| event charge        | 10 A decaying to 0 over 20 $\mu\text{s}$ = 100 $\mu\text{C}$                       |
|---------------------|------------------------------------------------------------------------------------|
| on 47 $\mu\text{F}$ | 2.14 V of rail rise                                                                |
| above setpoint      | charger is <b>off</b> , so the whole 100 V / 22 k $\Omega$ = 4.55 mA discharges it |
| time to clear       | <b>21.4 ms</b> , against <b>26.67 ms</b> between cylinders at 1500 rpm             |
| cost                | <b>455 mW</b> , burned continuously whenever the rail is up                        |

Checked by `boost_converter.cir` arrangement C, which runs two identical reservoirs — one bare, one bled — through three such events at the real cylinder spacing. Bare, the rail stacks to 106.4 V and stays there.

A comparator-gated shunt costs nothing at idle and is the alternative. It is more parts, and it is a part nobody has chosen.

Over a complete injection event there is no pumping at all — the peak phase draws 342  $\mu\text{C}$  from the reservoir and end-of-injection recirculation returns roughly 100  $\mu\text{C}$  of it. This resistor is not there for normal running. It is there because one-sided regulation has no answer for an excursion from *any* cause.

**INJ-RECIRC — injector recirculation diodes, five off**

150 V, 20 A, ultrafast. Three on the low sides (one per cylinder, into the boost reservoir) and two on the high sides (the battery diode-OR).

Research memo 10 names the class. Not a Zener active clamp across the FET — Nexperia AN50003 rates repetitive active clamp reliability- "Questionable" for gate-oxide wear-out. Not a bare freewheel to battery, which is too slow: the same problem the boost rail exists to fix on the opening side.

Sized by the pulse, not the average. `injector_turnoff.cir` measures 28.7 W instantaneous during turn-off against roughly 3.6 W averaged over all nine events per second — a factor of eight. The average would never have sized this part.

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## Chosen parts

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These are the entries that have moved from "a class" to "a part number", each against a datasheet that was actually retrieved and read. They are listed separately from the requirements above because they are answers, not constraints.

| Function           | Part             | Why this one                                                                                                |
|--------------------|------------------|-------------------------------------------------------------------------------------------------------------|
| Buck pre-regulator | LM5164DDAR       | 100 V input, 1 A, 6–100 V range. See below — it replaces a 60 V part that can no longer survive this input. |
| 3V3 regulator      | TLV76733QWDRBRQ1 | 1% over load <i>and</i> temperature. The supervisor makes that load-bearing.                                |
| Supervisor         | TPS3850G33DRCT   | $\pm 4\%$ window variant, so its undervoltage trip clears the MCU's own LVD by 143 mV rather than 44 mV.    |



## LM5164DDAR — and the 60 V part it replaces

Research memo 07 §4 selected **TPS54360B-Q1**, reasoning that a 60 V part beat a 42 V LM5175-Q1 "for load-dump margin". That reasoning was correct when it was written. It is not correct now, and the change came from the input-protection work above:

- The TVS standoff had to rise to 43 V so the part stops conducting during a *normal* clamped load dump.
- A higher standoff clamps higher. Pulse 2a now clamps at 73.3 V.
- `transient_clamp.cir` carries a **passing** check that says so in as many words: *"60 V-class parts are NO LONGER viable downstream."*

A 60 V buck on this input is a part that fails the first ISO 7637-2 pulse 2a event. The LM5164 was already the part every rating check in the suite was written against — its  $-0.3\text{ V} / 100\text{ V}$  VIN limits are what `negative_pulses` and `transient_clamp` check — but nothing had ever reconciled the BOM table with it. **This is what a contradiction between two documents looks like when only one of them is executable.**

Operating range 6–100 V covers the spec's own 6–40 V requirement, with the 6 V end matching the cranking dip the input range was built around. 1 A against an estimated 400–500 mA board load.

Configuration values, all derived from the datasheet rather than picked:

| Part            | Value                                                   | From                                                                                                                                                                                                                                                                                                                                            |
|-----------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RON             | 31.6 k $\Omega$                                         | The tON table in §5.5 is linear in RON/VIN: at 12 V, 25 k $\Omega$ gives 830 ns, so $K = 398\text{ ns}\cdot\text{V}/\text{k}\Omega$ . For 400 kHz at 5 V out, $\text{RON} = \text{tON}\cdot\text{VIN}/K$ is 31.4 k $\Omega$ at <b>both</b> 13.5 V and 40 V — the VIN feedforward is what holds the frequency fixed. E96 value 31.6 k $\Omega$ . |
| FB divider      | 38.3 k $\Omega$ / 12.1 k $\Omega$                       | VREF is 1.2 V (1.181–1.218, $\pm 1.5\%$ ), so 5 V needs $1 + R1/R2 = 4.167$ .                                                                                                                                                                                                                                                                   |
| EN/UVLO divider | 75 k $\Omega$ / 24.9 k $\Omega$                         | Enable rises at 1.5 V typ; 0.249 ratio puts turn-on at 6.02 V, the spec's cranking-dip floor. The pin is rated to 100 V, so the divider programs the threshold rather than protecting the pin.                                                                                                                                                  |
| BST cap         | 2.2 nF 50 V X7R                                         | Not a range. The datasheet specifies the part: <i>"a high-quality 2.2nF 50V X7R ceramic capacitor between BST and SW"</i> .                                                                                                                                                                                                                     |
| L / Cout        | 33 $\mu\text{H}$ / 47 $\mu\text{F}$ , 10 m $\Omega$ ESR | <code>buck_preregulator.cir</code> , sized at the 40 V worst case rather than nominal. The ESR is part of the value — it sets the ripple.                                                                                                                                                                                                       |

The RON figure is worth one more line: it independently reproduces `buck_preregulator.cir`'s own gate timings, 925 ns at 13.5 V and 312.5 ns at 40 V. The netlist got those from  $D = \text{Vout}/\text{Vin}$ ; the datasheet gets them from RON. They agree.

## TLV76733QWDRBQ1 — accuracy as a requirement

Memo 07 §6 named `TLV1117-33QDCYRQ1`, explicitly *"class pricing, not fetched"*. Choosing the supervisor turned 3V3 accuracy into a hard constraint: outside 3.143–3.459 V the board resets itself. TLV767-Q1 is



1% over load and temperature — 3.267–3.333 V, about 125 mV of margin at each end. 1 A, 2.5–16 V in, AEC-Q100, VSON-8.

The fixed-output version is the one specified. Its pin 3 is `SNS` where the adjustable version has `FB`, and they are not interchangeable. `SNS` must not float; it is tied to `OUT`.

### TPS3850G33DRCT

See `docs/pinmap.md` §1.8 and `hw/gen_symbols.py`. The variant letter is the decision: `G` sets thresholds at  $\pm 4\%$  of nominal and `H` at  $\pm 7\%$ , and with the part's  $\pm 0.8\%$  accuracy that is a worst-case undervoltage trip of 3.143 V against 3.044 V. The supervisor has to assert before the S32K148's own LVD (3.0 V max), so G33 clears it by 143 mV and H33 by 44 mV — close enough that the two could fire in either order, and a supervisor that might lose the race to the thing it supervises is not doing its job. The cost is the tighter rail window above.

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## MCU

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NXP S32K148, `FS32K148HAT0MLQT` — 144-pin LQFP, 80 MHz,  $-40\dots+125\text{ }^{\circ}\text{C}$ .

Confirmed in research memo 05 against the S32K1xx data sheet Rev. 15 with page-level citations. The full reasoning is there; what matters for the BOM:

- The 144-pin package is headroom against unknown U1 (50 of the OEM ECU's 94 pins are still unresolved), not a requirement of the confirmed channel budget, which fits inside a 100-pin part.
- **LQFP over MAPBGA:** identical instance counts — CAN/ADC/FTM/I<sup>2</sup>C are silicon-level, not package-level — and 0.5 mm pitch is inspectable by eye or AOI, where a BGA needs X-ray or destructive inspection. A real cost for a small team doing early assembly.
- **80 MHz H grade, not 112 MHz U:** the `U` speed grade is *not valid* at the 125 °C `M` temperature grade, per the datasheet's own ordering-table footnote. Memo 05 §6 shows the extra clock is not needed to close the timing budget, so there is no reason to trade thermal margin for headroom the design does not use. This corrects the spec §5 framing, which cites 112 MHz.
- In stock at DigiKey's India storefront at single-unit quantities. Not a lead-time part. The S32K146 fallback was checked and is *not* needed — it is the one that is out of stock, with 12–52 week lead times.

**Still to check before footprint lock:** package drawing 98ASS23177W (SOT486-2) has not been independently fetched. The PCB roadmap's warning stands — a hand-sourced footprint is the most likely place to introduce a silent error.